# Quotient Space

Prop. 1 : $\pi_W : V \to V/W$,

$\pi_W(\underline{v}) := \underline{v} + W$

$\pi_W$ is linear (1) & surj. (2) & $\ker(\pi_W) = W$ (3)

Proof :

(1) $\pi_W(\underline{v}+\underline{v}') = (\underline{v}+\underline{v}')W$

[v. space add. in $V/W$] $= (\underline{v}+W) + (\underline{v}'+W$

$= \pi_W(\underline{v}) + \pi_W(\underline{v}')$

Ex. : 証 $\pi_W(\alpha\underline{v}) = \alpha\pi_W(\underline{v})$

(2) Obviously

(3) $\pi_W(\underline{v}) = \underline{0}_{V/W} = \underline{0} + W$

$\Leftrightarrow \underline{v} + W = \underline{0} + W$

(Theorem) $\Leftrightarrow \underline{v} - \underline{0} \in W$

Prop. 2: If $\dim(V) < \infty$,
$W \leq V \Rightarrow \dim(V/W) = \dim(V) - \dim(W)$

Proof:

Apply rank-nullity Theorem on $\pi_W$

① $\dim(V) = \dim(\underbrace{\ker(\pi_W)}_{=W}) + \dim(\underbrace{\mathrm{Im}(\pi_W)}_{=V/W})$

$\bar{T}$ Def.: $T: V \to W$

Suppose $V' \leq \ker(T)$

$\bar{T}: V/V' \to W$

$\bar{T}(\underline{v}+V') := T(\underline{v})$

(old l. trans.)

↓ (define)

(new l. trans.)

Prop. 3:

(1) $\bar{T}$ is well-defined,

i.e., suppose $\underline{v}+V' = \underline{u}+V'$ $(\underline{v} \neq \underline{u})$

then $\bar{T}(\underline{v}+V') = \bar{T}(\underline{u}+V') \Leftrightarrow T(\underline{v}) = T(\underline{u})$

(WTS)

Proof: $T(\underline{v}) - T(\underline{u}) = T(\underline{v}-\underline{u})$ $(\underline{v}-\underline{u} \in V')$

$= T(\underline{v}')$ $(\underline{v}' \in V')$

$= \underline{0}$ (② $V' \leq \ker(T)$)

$\Rightarrow T(\underline{v}) = T(\underline{u})$ ②

(2) $\bar{T}$ is a linear transformation

Proof: $\bar{T}((\underline{v}+V')+(\underline{u}+V'))$

$= \bar{T}((\underline{v}+\underline{u})+V')$

$= T(\underline{v}+\underline{u})$

(⊗ Def. of $\bar{T}$) → $= T(\underline{v}) + T(\underline{u})$

$= \bar{T}(\underline{v}+V') + \bar{T}(\underline{u}+V')$

(HW: 証: if we take $V' = \ker(T)$,

then $\bar{T}: V/\ker(T) \to W$ is injective)

# Dual Space $V^*$

Def.: $V^* := \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F})$ ← f-dim. v. space

All elements $\phi$ in $V^*$ are called linear functionals

Recall:

① $\mathrm{Hom}_{\mathbb{F}}(V, W)$ is a vector space

(及HW4) Suppose $\phi, \psi \in \mathrm{Hom}_{\mathbb{F}}(V, W)$,

then $(\phi+\psi), \alpha\phi \in \mathrm{Hom}_{\mathbb{F}}(V, W)$

are defined by:

$$(\phi+\psi)(\underline{v}) := \phi(\underline{v}) + \psi(\underline{v})$$

$$(\alpha\phi)(\underline{v}) := \alpha\phi(\underline{v})$$

② $\forall$ l. trans $\phi: V \to W$,

if $B := \{\underline{v}_i \mid i \in I\}$ is a basis of $V$,

then $\phi$ is uniquely determined

by the value of $\phi(\underline{v}_i)$ for all $i \in I$

e.g. For $T: \mathbb{R}^n \to \mathbb{R}^m$,

if we know $T(\underline{e}_i)$ for $i = 1, \cdots, n$

then $T(\underline{v}) = T(\sum_{i=1}^{n} \alpha_i \underline{e}_i) = \sum_{i=1}^{n} \alpha_i T(e_i)$

②

E.g. (1) $V = M_{n\times n}(\mathbb{F})$,

define $\phi: V \to \mathbb{F}$ by $\phi(A) := A_{11}$

Claim: $\phi \in V^* = \mathrm{Hom}(V, \mathbb{F})$

(見證)

(2) $V = M_{n\times n}(\mathbb{F})$, $tr: V \to \mathbb{F}$

(trace) $tr(A) := A_{11} + A_{22} + \cdots + A_{nn}$

Then $tr \in V^*$

(3) $V = M_{n\times n}(\mathbb{F})$, $\det: V \to \mathbb{F}$

Then for $n > 1$, $\det \notin V^*$

since $\det(A+B) \neq \det(A) + \det(B)$ in general

Also, $\det(\alpha A) = \alpha^n \det(A) \neq \alpha \det(A)$

(4) $V = C^\infty(\mathbb{R})$, $\phi: V \to \mathbb{F}$, $\phi(f) := f''(1)$

Then $\phi \in V^*$

e.g. $\phi(f+g) = (f+g)''(1) = f''(1) + g''(1)$

$= \phi(f) + \phi(g)$

Def.: Let $B_i = \{v_i \mid i \in I\}$ be a basis of $V$

can be infinite

For each $i \in I$,

define $\phi_i : V \to \mathbb{F}$ by:

Step 1: On $B$,

$$\phi_i(v_j) = \delta_{ij} := \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

Step 2: On $V$

for any $v = \alpha_1 v_{i_1} + \cdots + \alpha_l v_{i_l} \in V$

$(v_{i_k} \in B)$,

$\phi_i(v) :=$

(Ex.) → Then by construction, $\phi_i \in V^*$

Define $B^* = \{\phi_i \mid i \in I\} \in V^*$

Q: Is $B^*$ a basis of $V^*$?

E.g.: $V = \mathbb{F}^n$, $B = \{e_i \mid 1 \le i \le n\}$